# MSG No. 229.140 $Im\bar{3}m$ [ Type I, cubic ]

Table 1: Wyckoff site: `2a`, site symmetry: `m-3m`

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | [1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24, |
| | | 25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72, |
| | | 73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96] |

Table 2: Wyckoff site: `6b`, site symmetry: `4/mm.m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [1,2,3,8,9,10,13,14,25,26,27,32,33,34,37,38] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [4,5,15,16,18,19,21,23,28,29,39,40,42,43,45,47] |
| 3 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [6,7,11,12,17,20,22,24,30,31,35,36,41,44,46,48] |
| 4 | $\left[\frac{1}{2}, 0, 0\right]$ | [49,50,51,56,57,58,61,62,73,74,75,80,81,82,85,86] |
| 5 | $\left[0, 0, \frac{1}{2}\right]$ | [52,53,63,64,66,67,69,71,76,77,87,88,90,91,93,95] |
| 6 | $\left[0, \frac{1}{2}, 0\right]$ | [54,55,59,60,65,68,70,72,78,79,83,84,89,92,94,96] |

Table 3: Wyckoff site: `8c`, site symmetry: `.-3m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [1,17,18,36,38,40,60,62,64,73,89,90] |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [2,7,15,33,43,48,57,67,72,74,79,87] |
| 3 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [3,4,11,34,44,45,58,68,69,75,76,83] |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [5,6,13,32,46,47,56,70,71,77,78,85] |
| 5 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [8,22,23,29,30,37,53,54,61,80,94,95] |
| 6 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [9,19,24,26,31,39,50,55,63,81,91,96] |
| 7 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [10,20,21,27,28,35,51,52,59,82,92,93] |
| 8 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [12,14,16,25,41,42,49,65,66,84,86,88] |

Table 4: Wyckoff site: `12d`, site symmetry: `-4m.2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, 0, \frac{1}{2}\right]$ | [1,8,33,34,61,62,74,75] |
| 2 | $\left[\frac{1}{4}, \frac{1}{2}, 0\right]$ | [2,3,37,38,57,58,73,80] |
| 3 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | [4,16,29,39,66,69,91,95] |
| 4 | $\left[\frac{1}{2}, 0, \frac{1}{4}\right]$ | [5,15,28,40,67,71,90,93] |
| 5 | $\left[0, \frac{1}{4}, \frac{1}{2}\right]$ | [6,11,31,36,68,70,89,96] |
| 6 | $\left[0, \frac{3}{4}, \frac{1}{2}\right]$ | [7,12,30,35,65,72,92,94] |
| 7 | $\left[\frac{3}{4}, 0, \frac{1}{2}\right]$ | [9,10,25,32,50,51,85,86] |
| 8 | $\left[\frac{3}{4}, \frac{1}{2}, 0\right]$ | [13,14,26,27,49,56,81,82] |
| 9 | $\left[\frac{1}{2}, \frac{1}{4}, 0\right]$ | [17,24,44,46,55,60,78,83] |
| 10 | $\left[0, \frac{1}{2}, \frac{1}{4}\right]$ | [18,21,43,47,52,64,77,87] |
| 11 | $\left[0, \frac{1}{2}, \frac{3}{4}\right]$ | [19,23,42,45,53,63,76,88] |

*continued ...*

Table 4

| No. | position | mapping |
|---|---|---|
| 12 | $\left[\frac{1}{2}, \frac{3}{4}, 0\right]$ | `[20,22,41,48,54,59,79,84]` |

Table 5: Wyckoff site: `12e`, site symmetry: `4m.m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, 0]$ | `[1,2,3,8,33,34,37,38]` |
| 2 | $[0, 0, -x]$ | `[4,16,19,23,29,39,42,45]` |
| 3 | $[0, 0, x]$ | `[5,15,18,21,28,40,43,47]` |
| 4 | $[0, x, 0]$ | `[6,11,17,24,31,36,44,46]` |
| 5 | $[0, -x, 0]$ | `[7,12,20,22,30,35,41,48]` |
| 6 | $[-x, 0, 0]$ | `[9,10,13,14,25,26,27,32]` |
| 7 | $\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[49,50,51,56,81,82,85,86]` |
| 8 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$ | `[52,64,67,71,77,87,90,93]` |
| 9 | $\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$ | `[53,63,66,69,76,88,91,95]` |
| 10 | $\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$ | `[54,59,65,72,79,84,92,94]` |
| 11 | $\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$ | `[55,60,68,70,78,83,89,96]` |
| 12 | $\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$ | `[57,58,61,62,73,74,75,80]` |

Table 6: Wyckoff site: `16f`, site symmetry: `.3m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, x]$ | `[1,17,18,36,38,40]` |
| 2 | $[x, -x, x]$ | `[2,7,15,33,43,48]` |
| 3 | $[x, x, -x]$ | `[3,4,11,34,44,45]` |
| 4 | $[-x, x, x]$ | `[5,6,13,32,46,47]` |
| 5 | $[x, -x, -x]$ | `[8,22,23,29,30,37]` |
| 6 | $[-x, x, -x]$ | `[9,19,24,26,31,39]` |
| 7 | $[-x, -x, x]$ | `[10,20,21,27,28,35]` |
| 8 | $[-x, -x, -x]$ | `[12,14,16,25,41,42]` |
| 9 | $\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$ | `[49,65,66,84,86,88]` |
| 10 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, x+\frac{1}{2}\right]$ | `[50,55,63,81,91,96]` |
| 11 | $\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-x\right]$ | `[51,52,59,82,92,93]` |
| 12 | $\left[\frac{1}{2}-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$ | `[53,54,61,80,94,95]` |
| 13 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$ | `[56,70,71,77,78,85]` |
| 14 | $\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$ | `[57,67,72,74,79,87]` |
| 15 | $\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$ | `[58,68,69,75,76,83]` |
| 16 | $\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$ | `[60,62,64,73,89,90]` |

Table 7: Wyckoff site: `24g`, site symmetry: `2mm..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, 0, \frac{1}{2}\right]$ | `[1,8,33,34]` |
| 2 | $\left[x, \frac{1}{2}, 0\right]$ | `[2,3,37,38]` |

*continued ...*

Table 7

| No. | position | mapping |
|---|---|---|
| 3 | $\left[\frac{1}{2}, 0, -x\right]$ | [4,16,29,39] |
| 4 | $\left[\frac{1}{2}, 0, x\right]$ | [5,15,28,40] |
| 5 | $\left[0, x, \frac{1}{2}\right]$ | [6,11,31,36] |
| 6 | $\left[0, -x, \frac{1}{2}\right]$ | [7,12,30,35] |
| 7 | $\left[-x, 0, \frac{1}{2}\right]$ | [9,10,25,32] |
| 8 | $\left[-x, \frac{1}{2}, 0\right]$ | [13,14,26,27] |
| 9 | $\left[\frac{1}{2}, x, 0\right]$ | [17,24,44,46] |
| 10 | $\left[0, \frac{1}{2}, x\right]$ | [18,21,43,47] |
| 11 | $\left[0, \frac{1}{2}, -x\right]$ | [19,23,42,45] |
| 12 | $\left[\frac{1}{2}, -x, 0\right]$ | [20,22,41,48] |
| 13 | $\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$ | [49,56,81,82] |
| 14 | $\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$ | [50,51,85,86] |
| 15 | $\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$ | [52,64,77,87] |
| 16 | $\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$ | [53,63,76,88] |
| 17 | $\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$ | [54,59,79,84] |
| 18 | $\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$ | [55,60,78,83] |
| 19 | $\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$ | [57,58,73,80] |
| 20 | $\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$ | [61,62,74,75] |
| 21 | $\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$ | [65,72,92,94] |
| 22 | $\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$ | [66,69,91,95] |
| 23 | $\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$ | [67,71,90,93] |
| 24 | $\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$ | [68,70,89,96] |

Table 8: Wyckoff site: 24h, site symmetry: m.2m

| No. | position | mapping |
|---|---|---|
| 1 | $[0, y, y]$ | [1,13,32,38] |
| 2 | $[0, -y, y]$ | [2,10,27,33] |
| 3 | $[0, y, -y]$ | [3,9,26,34] |
| 4 | $[y, y, 0]$ | [4,18,40,45] |
| 5 | $[-y, y, 0]$ | [5,19,39,47] |
| 6 | $[-y, 0, y]$ | [6,20,35,46] |
| 7 | $[y, 0, y]$ | [7,17,36,48] |
| 8 | $[0, -y, -y]$ | [8,14,25,37] |
| 9 | $[y, 0, -y]$ | [11,22,30,44] |
| 10 | $[-y, 0, -y]$ | [12,24,31,41] |
| 11 | $[y, -y, 0]$ | [15,23,29,43] |
| 12 | $[-y, -y, 0]$ | [16,21,28,42] |
| 13 | $\left[\frac{1}{2}, y+\frac{1}{2}, y+\frac{1}{2}\right]$ | [49,61,80,86] |
| 14 | $\left[\frac{1}{2}, \frac{1}{2}-y, y+\frac{1}{2}\right]$ | [50,58,75,81] |
| 15 | $\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-y\right]$ | [51,57,74,82] |
| 16 | $\left[y+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}\right]$ | [52,66,88,93] |
| 17 | $\left[\frac{1}{2}-y, y+\frac{1}{2}, \frac{1}{2}\right]$ | [53,67,87,95] |
| 18 | $\left[\frac{1}{2}-y, \frac{1}{2}, y+\frac{1}{2}\right]$ | [54,68,83,94] |
| 19 | $\left[y+\frac{1}{2}, \frac{1}{2}, y+\frac{1}{2}\right]$ | [55,65,84,96] |

*continued ...*

Table 8

| No. | position | mapping |
|---|---|---|
| 20 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-y\right]$ | [56,62,73,85] |
| 21 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-y\right]$ | [59,70,78,92] |
| 22 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-y\right]$ | [60,72,79,89] |
| 23 | $\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [63,71,77,91] |
| 24 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [64,69,76,90] |

Table 9: Wyckoff site: 48i, site symmetry: ..2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$ | [1,62] |
| 2 | $\left[\frac{1}{4},\ y+\frac{1}{2},\ y\right]$ | [2,57] |
| 3 | $\left[\frac{1}{4},\ \frac{1}{2}-y,\ -y\right]$ | [3,58] |
| 4 | $\left[\frac{1}{2}-y,\ y,\ \frac{3}{4}\right]$ | [4,69] |
| 5 | $\left[y+\frac{1}{2},\ y,\ \frac{1}{4}\right]$ | [5,71] |
| 6 | $\left[-y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$ | [6,70] |
| 7 | $\left[y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$ | [7,72] |
| 8 | $\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$ | [8,61] |
| 9 | $\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$ | [9,50] |
| 10 | $\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$ | [10,51] |
| 11 | $\left[y,\ \frac{1}{4},\ y+\frac{1}{2}\right]$ | [11,68] |
| 12 | $\left[-y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$ | [12,65] |
| 13 | $\left[\frac{3}{4},\ \frac{1}{2}-y,\ y\right]$ | [13,56] |
| 14 | $\left[\frac{3}{4},\ y+\frac{1}{2},\ -y\right]$ | [14,49] |
| 15 | $\left[\frac{1}{2}-y,\ -y,\ \frac{1}{4}\right]$ | [15,67] |
| 16 | $\left[y+\frac{1}{2},\ -y,\ \frac{3}{4}\right]$ | [16,66] |
| 17 | $\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$ | [17,60] |
| 18 | $\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$ | [18,64] |
| 19 | $\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$ | [19,63] |
| 20 | $\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$ | [20,59] |
| 21 | $\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$ | [21,52] |
| 22 | $\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$ | [22,54] |
| 23 | $\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$ | [23,53] |
| 24 | $\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$ | [24,55] |
| 25 | $\left[\frac{3}{4},\ -y,\ y+\frac{1}{2}\right]$ | [25,86] |
| 26 | $\left[\frac{3}{4},\ \frac{1}{2}-y,\ -y\right]$ | [26,81] |
| 27 | $\left[\frac{3}{4},\ y+\frac{1}{2},\ y\right]$ | [27,82] |
| 28 | $\left[y+\frac{1}{2},\ -y,\ \frac{1}{4}\right]$ | [28,93] |
| 29 | $\left[\frac{1}{2}-y,\ -y,\ \frac{3}{4}\right]$ | [29,95] |
| 30 | $\left[y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$ | [30,94] |
| 31 | $\left[-y,\ \frac{1}{4},\ y+\frac{1}{2}\right]$ | [31,96] |
| 32 | $\left[\frac{3}{4},\ y,\ \frac{1}{2}-y\right]$ | [32,85] |
| 33 | $\left[\frac{1}{4},\ -y,\ \frac{1}{2}-y\right]$ | [33,74] |
| 34 | $\left[\frac{1}{4},\ y,\ y+\frac{1}{2}\right]$ | [34,75] |
| 35 | $\left[-y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$ | [35,92] |
| 36 | $\left[y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$ | [36,89] |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 37 | $\left[\frac{1}{4}, y+\frac{1}{2}, -y\right]$ | [37,80] |
| 38 | $\left[\frac{1}{4}, \frac{1}{2}-y, y\right]$ | [38,73] |
| 39 | $\left[y+\frac{1}{2}, y, \frac{3}{4}\right]$ | [39,91] |
| 40 | $\left[\frac{1}{2}-y, y, \frac{1}{4}\right]$ | [40,90] |
| 41 | $\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$ | [41,84] |
| 42 | $\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$ | [42,88] |
| 43 | $\left[y, y+\frac{1}{2}, \frac{1}{4}\right]$ | [43,87] |
| 44 | $\left[\frac{1}{2}-y, \frac{1}{4}, -y\right]$ | [44,83] |
| 45 | $\left[y, \frac{1}{2}-y, \frac{3}{4}\right]$ | [45,76] |
| 46 | $\left[y+\frac{1}{2}, \frac{1}{4}, y\right]$ | [46,78] |
| 47 | $\left[-y, \frac{1}{2}-y, \frac{1}{4}\right]$ | [47,77] |
| 48 | $\left[\frac{1}{2}-y, \frac{3}{4}, y\right]$ | [48,79] |

Table 10: Wyckoff site: 48j, site symmetry: m..

| No. | position | mapping |
|---|---|---|
| 1 | $[0, y, z]$ | [1,32] |
| 2 | $[0, -z, y]$ | [2,27] |
| 3 | $[0, z, -y]$ | [3,26] |
| 4 | $[z, y, 0]$ | [4,40] |
| 5 | $[-z, y, 0]$ | [5,39] |
| 6 | $[-y, 0, z]$ | [6,35] |
| 7 | $[y, 0, z]$ | [7,36] |
| 8 | $[0, -y, -z]$ | [8,25] |
| 9 | $[0, y, -z]$ | [9,34] |
| 10 | $[0, -y, z]$ | [10,33] |
| 11 | $[y, 0, -z]$ | [11,30] |
| 12 | $[-y, 0, -z]$ | [12,31] |
| 13 | $[0, z, y]$ | [13,38] |
| 14 | $[0, -z, -y]$ | [14,37] |
| 15 | $[z, -y, 0]$ | [15,29] |
| 16 | $[-z, -y, 0]$ | [16,28] |
| 17 | $[z, 0, y]$ | [17,48] |
| 18 | $[y, z, 0]$ | [18,45] |
| 19 | $[-y, z, 0]$ | [19,47] |
| 20 | $[-z, 0, y]$ | [20,46] |
| 21 | $[-y, -z, 0]$ | [21,42] |
| 22 | $[z, 0, -y]$ | [22,44] |
| 23 | $[y, -z, 0]$ | [23,43] |
| 24 | $[-z, 0, -y]$ | [24,41] |
| 25 | $\left[\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$ | [49,80] |
| 26 | $\left[\frac{1}{2}, \frac{1}{2}-z, y+\frac{1}{2}\right]$ | [50,75] |
| 27 | $\left[\frac{1}{2}, z+\frac{1}{2}, \frac{1}{2}-y\right]$ | [51,74] |
| 28 | $\left[z+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}\right]$ | [52,88] |
| 29 | $\left[\frac{1}{2}-z, y+\frac{1}{2}, \frac{1}{2}\right]$ | [53,87] |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 30 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | [54,83] |
| 31 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | [55,84] |
| 32 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | [56,73] |
| 33 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [57,82] |
| 34 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [58,81] |
| 35 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | [59,78] |
| 36 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | [60,79] |
| 37 | $\left[\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$ | [61,86] |
| 38 | $\left[\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$ | [62,85] |
| 39 | $\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [63,77] |
| 40 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [64,76] |
| 41 | $\left[z+\frac{1}{2},\ \frac{1}{2},\ y+\frac{1}{2}\right]$ | [65,96] |
| 42 | $\left[y+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}\right]$ | [66,93] |
| 43 | $\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ \frac{1}{2}\right]$ | [67,95] |
| 44 | $\left[\frac{1}{2}-z,\ \frac{1}{2},\ y+\frac{1}{2}\right]$ | [68,94] |
| 45 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ \frac{1}{2}\right]$ | [69,90] |
| 46 | $\left[z+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-y\right]$ | [70,92] |
| 47 | $\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}\right]$ | [71,91] |
| 48 | $\left[\frac{1}{2}-z,\ \frac{1}{2},\ \frac{1}{2}-y\right]$ | [72,89] |

Table 11: Wyckoff site: 48k, site symmetry: ..m

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ z]$ | [1,36] |
| 2 | $[x,\ -z,\ x]$ | [2,43] |
| 3 | $[x,\ z,\ -x]$ | [3,45] |
| 4 | $[z,\ x,\ -x]$ | [4,44] |
| 5 | $[-z,\ x,\ x]$ | [5,46] |
| 6 | $[-x,\ x,\ z]$ | [6,32] |
| 7 | $[x,\ -x,\ z]$ | [7,33] |
| 8 | $[x,\ -x,\ -z]$ | [8,30] |
| 9 | $[-x,\ x,\ -z]$ | [9,31] |
| 10 | $[-x,\ -x,\ z]$ | [10,35] |
| 11 | $[x,\ x,\ -z]$ | [11,34] |
| 12 | $[-x,\ -x,\ -z]$ | [12,25] |
| 13 | $[-x,\ z,\ x]$ | [13,47] |
| 14 | $[-x,\ -z,\ -x]$ | [14,42] |
| 15 | $[z,\ -x,\ x]$ | [15,48] |
| 16 | $[-z,\ -x,\ -x]$ | [16,41] |
| 17 | $[z,\ x,\ x]$ | [17,40] |
| 18 | $[x,\ z,\ x]$ | [18,38] |
| 19 | $[-x,\ z,\ -x]$ | [19,26] |
| 20 | $[-z,\ -x,\ x]$ | [20,28] |
| 21 | $[-x,\ -z,\ x]$ | [21,27] |
| 22 | $[z,\ -x,\ -x]$ | [22,29] |

*continued ...*

Table 11

| No. | position | mapping |
|---|---|---|
| 23 | $[x, -z, -x]$ | `[23,37]` |
| 24 | $[-z, x, -x]$ | `[24,39]` |
| 25 | $\left[x+\frac{1}{2}, x+\frac{1}{2}, z+\frac{1}{2}\right]$ | `[49,84]` |
| 26 | $\left[x+\frac{1}{2}, \frac{1}{2}-z, x+\frac{1}{2}\right]$ | `[50,91]` |
| 27 | $\left[x+\frac{1}{2}, z+\frac{1}{2}, \frac{1}{2}-x\right]$ | `[51,93]` |
| 28 | $\left[z+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-x\right]$ | `[52,92]` |
| 29 | $\left[\frac{1}{2}-z, x+\frac{1}{2}, x+\frac{1}{2}\right]$ | `[53,94]` |
| 30 | $\left[\frac{1}{2}-x, x+\frac{1}{2}, z+\frac{1}{2}\right]$ | `[54,80]` |
| 31 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, z+\frac{1}{2}\right]$ | `[55,81]` |
| 32 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-z\right]$ | `[56,78]` |
| 33 | $\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$ | `[57,79]` |
| 34 | $\left[\frac{1}{2}-x, \frac{1}{2}-x, z+\frac{1}{2}\right]$ | `[58,83]` |
| 35 | $\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-z\right]$ | `[59,82]` |
| 36 | $\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-z\right]$ | `[60,73]` |
| 37 | $\left[\frac{1}{2}-x, z+\frac{1}{2}, x+\frac{1}{2}\right]$ | `[61,95]` |
| 38 | $\left[\frac{1}{2}-x, \frac{1}{2}-z, \frac{1}{2}-x\right]$ | `[62,90]` |
| 39 | $\left[z+\frac{1}{2}, \frac{1}{2}-x, x+\frac{1}{2}\right]$ | `[63,96]` |
| 40 | $\left[\frac{1}{2}-z, \frac{1}{2}-x, \frac{1}{2}-x\right]$ | `[64,89]` |
| 41 | $\left[z+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$ | `[65,88]` |
| 42 | $\left[x+\frac{1}{2}, z+\frac{1}{2}, x+\frac{1}{2}\right]$ | `[66,86]` |
| 43 | $\left[\frac{1}{2}-x, z+\frac{1}{2}, \frac{1}{2}-x\right]$ | `[67,74]` |
| 44 | $\left[\frac{1}{2}-z, \frac{1}{2}-x, x+\frac{1}{2}\right]$ | `[68,76]` |
| 45 | $\left[\frac{1}{2}-x, \frac{1}{2}-z, x+\frac{1}{2}\right]$ | `[69,75]` |
| 46 | $\left[z+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$ | `[70,77]` |
| 47 | $\left[x+\frac{1}{2}, \frac{1}{2}-z, \frac{1}{2}-x\right]$ | `[71,85]` |
| 48 | $\left[\frac{1}{2}-z, x+\frac{1}{2}, \frac{1}{2}-x\right]$ | `[72,87]` |

Table 12: Wyckoff site: `96l`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, z]$ | `[1]` |
| 2 | $[x, -z, y]$ | `[2]` |
| 3 | $[x, z, -y]$ | `[3]` |
| 4 | $[z, y, -x]$ | `[4]` |
| 5 | $[-z, y, x]$ | `[5]` |
| 6 | $[-y, x, z]$ | `[6]` |
| 7 | $[y, -x, z]$ | `[7]` |
| 8 | $[x, -y, -z]$ | `[8]` |
| 9 | $[-x, y, -z]$ | `[9]` |
| 10 | $[-x, -y, z]$ | `[10]` |
| 11 | $[y, x, -z]$ | `[11]` |
| 12 | $[-y, -x, -z]$ | `[12]` |
| 13 | $[-x, z, y]$ | `[13]` |
| 14 | $[-x, -z, -y]$ | `[14]` |
| 15 | $[z, -y, x]$ | `[15]` |

*continued ...*

Table 12

| No. | position | mapping |
|---|---|---|
| 16 | $[-z, -y, -x]$ | [16] |
| 17 | $[z, x, y]$ | [17] |
| 18 | $[y, z, x]$ | [18] |
| 19 | $[-y, z, -x]$ | [19] |
| 20 | $[-z, -x, y]$ | [20] |
| 21 | $[-y, -z, x]$ | [21] |
| 22 | $[z, -x, -y]$ | [22] |
| 23 | $[y, -z, -x]$ | [23] |
| 24 | $[-z, x, -y]$ | [24] |
| 25 | $[-x, -y, -z]$ | [25] |
| 26 | $[-x, z, -y]$ | [26] |
| 27 | $[-x, -z, y]$ | [27] |
| 28 | $[-z, -y, x]$ | [28] |
| 29 | $[z, -y, -x]$ | [29] |
| 30 | $[y, -x, -z]$ | [30] |
| 31 | $[-y, x, -z]$ | [31] |
| 32 | $[-x, y, z]$ | [32] |
| 33 | $[x, -y, z]$ | [33] |
| 34 | $[x, y, -z]$ | [34] |
| 35 | $[-y, -x, z]$ | [35] |
| 36 | $[y, x, z]$ | [36] |
| 37 | $[x, -z, -y]$ | [37] |
| 38 | $[x, z, y]$ | [38] |
| 39 | $[-z, y, -x]$ | [39] |
| 40 | $[z, y, x]$ | [40] |
| 41 | $[-z, -x, -y]$ | [41] |
| 42 | $[-y, -z, -x]$ | [42] |
| 43 | $[y, -z, x]$ | [43] |
| 44 | $[z, x, -y]$ | [44] |
| 45 | $[y, z, -x]$ | [45] |
| 46 | $[-z, x, y]$ | [46] |
| 47 | $[-y, z, x]$ | [47] |
| 48 | $[z, -x, y]$ | [48] |
| 49 | $\left[x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$ | [49] |
| 50 | $\left[x+\frac{1}{2}, \frac{1}{2}-z, y+\frac{1}{2}\right]$ | [50] |
| 51 | $\left[x+\frac{1}{2}, z+\frac{1}{2}, \frac{1}{2}-y\right]$ | [51] |
| 52 | $\left[z+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-x\right]$ | [52] |
| 53 | $\left[\frac{1}{2}-z, y+\frac{1}{2}, x+\frac{1}{2}\right]$ | [53] |
| 54 | $\left[\frac{1}{2}-y, x+\frac{1}{2}, z+\frac{1}{2}\right]$ | [54] |
| 55 | $\left[y+\frac{1}{2}, \frac{1}{2}-x, z+\frac{1}{2}\right]$ | [55] |
| 56 | $\left[x+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}-z\right]$ | [56] |
| 57 | $\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$ | [57] |
| 58 | $\left[\frac{1}{2}-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$ | [58] |
| 59 | $\left[y+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-z\right]$ | [59] |
| 60 | $\left[\frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}-z\right]$ | [60] |
| 61 | $\left[\frac{1}{2}-x, z+\frac{1}{2}, y+\frac{1}{2}\right]$ | [61] |
| 62 | $\left[\frac{1}{2}-x, \frac{1}{2}-z, \frac{1}{2}-y\right]$ | [62] |

*continued ...*

Table 12

| No. | position | mapping |
|---|---|---|
| 63 | $\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$ | [63] |
| 64 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$ | [64] |
| 65 | $\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$ | [65] |
| 66 | $\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [66] |
| 67 | $\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [67] |
| 68 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y+\frac{1}{2}\right]$ | [68] |
| 69 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$ | [69] |
| 70 | $\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$ | [70] |
| 71 | $\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$ | [71] |
| 72 | $\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$ | [72] |
| 73 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | [73] |
| 74 | $\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$ | [74] |
| 75 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$ | [75] |
| 76 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$ | [76] |
| 77 | $\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$ | [77] |
| 78 | $\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [78] |
| 79 | $\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [79] |
| 80 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [80] |
| 81 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [81] |
| 82 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [82] |
| 83 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [83] |
| 84 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [84] |
| 85 | $\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$ | [85] |
| 86 | $\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$ | [86] |
| 87 | $\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [87] |
| 88 | $\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [88] |
| 89 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$ | [89] |
| 90 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$ | [90] |
| 91 | $\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$ | [91] |
| 92 | $\left[z+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$ | [92] |
| 93 | $\left[y+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [93] |
| 94 | $\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$ | [94] |
| 95 | $\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [95] |
| 96 | $\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ y+\frac{1}{2}\right]$ | [96] |